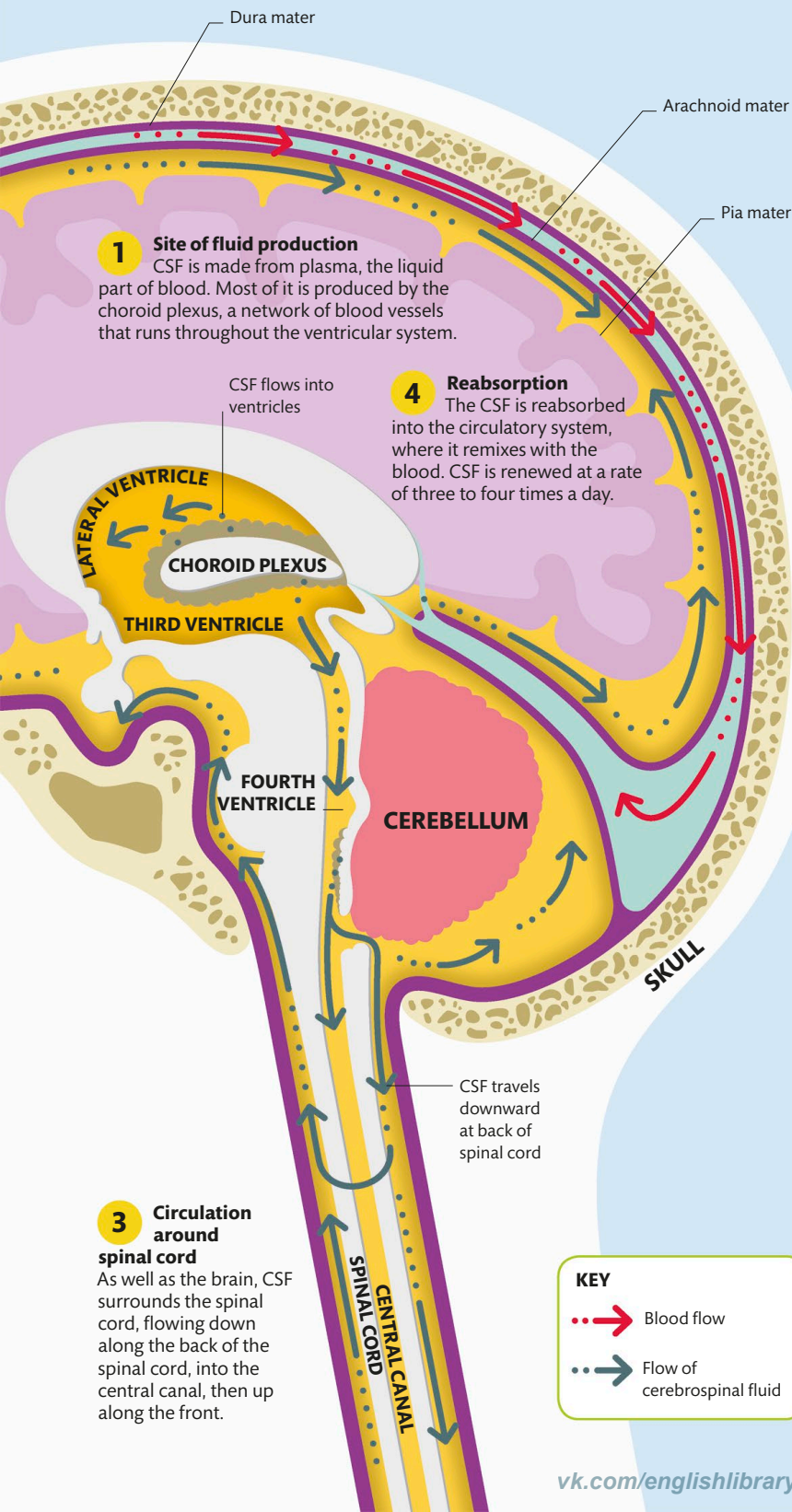
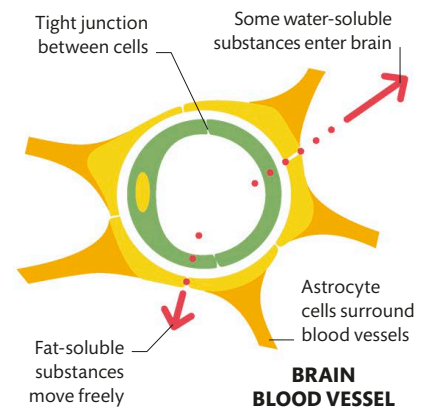
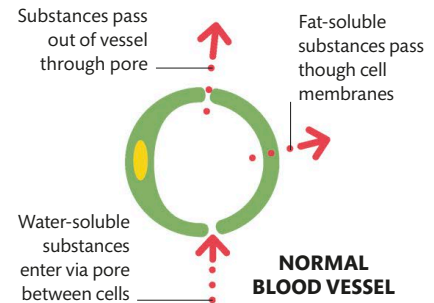




The blood-brain barrier

Infections from the rest of the body do not ordinarily reach the brain due to a system called the blood-brain barrier. As a general rule, blood capillaries in the rest of the body leak fluid easily (and any viruses and germs it contains) into surrounding tissues through gaps between the cells that form the blood vessel's wall. In the brain, these same cells have a much tighter fit, and the flow of materials between the brain is instead controlled by astrocytes that surround the blood vessels.



KEY

- Blood flow
- Flow of cerebrospinal fluid

Selectively permeable

Normal blood vessels allow fluid to pass through easily. However, while oxygen, fat-based hormones, and non-water-soluble materials pass through the blood-brain barrier unhindered, water-soluble items are blocked so they don't reach the CSF.